

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

```
df = pd.read_csv('/content/Invistico_Airline.csv')
```

```
df.head()
```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	...	Online support	Ease of Online booking	s
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0		0	0	2	...	2.0	3.0
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0		0	0	3	...	2.0	3.0
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0		0	0	3	...	2.0	2.0
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0		0	0	3	...	3.0	1.0
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0		0	0	3	...	4.0	2.0

5 rows × 22 columns

```
df.isna().sum()
```

	0
satisfaction	0
Customer Type	0
Age	0
Type of Travel	0
Class	0
Flight Distance	0
Seat comfort	0
Departure/Arrival time convenient	0
Food and drink	0
Gate location	0
Inflight wifi service	0
Inflight entertainment	0
Online support	1
Ease of Online booking	1
On-board service	1
Leg room service	1
Baggage handling	1
Checkin service	1
Cleanliness	1
Online boarding	1
Departure Delay in Minutes	1
Arrival Delay in Minutes	92

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23616 entries, 0 to 23615
Data columns (total 22 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   satisfaction                          23616 non-null  object
 1   Customer Type                        23616 non-null  object
 2   Age                                  23616 non-null  int64
 3   Type of Travel                      23616 non-null  object
 4   Class                               23616 non-null  object
 5   Flight Distance                     23616 non-null  int64
 6   Seat comfort                        23616 non-null  int64
 7   Departure/Arrival time convenient  23616 non-null  int64
 8   Food and drink                      23616 non-null  int64
 9   Gate location                       23616 non-null  int64
10  Inflight wifi service               23616 non-null  int64
11  Inflight entertainment              23616 non-null  int64
12  Online support                      23615 non-null  float64
13  Ease of Online booking              23615 non-null  float64
14  On-board service                    23615 non-null  float64
15  Leg room service                    23615 non-null  float64
16  Baggage handling                    23615 non-null  float64
17  Checkin service                     23615 non-null  float64
18  Cleanliness                         23615 non-null  float64
19  Online boarding                     23615 non-null  float64
20  Departure Delay in Minutes          23615 non-null  float64
21  Arrival Delay in Minutes            23524 non-null  float64
dtypes: float64(10), int64(8), object(4)
memory usage: 4.0+ MB
```

```
df['Arrival Delay in Minutes'].fillna(df['Arrival Delay in Minutes'].median(), inplace=True)
```

/tmp/ipykernel_15304/3226891278.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chainable .loc/.iloc. This behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting the values is a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df.method({col: value})

```
df['Arrival Delay in Minutes'].fillna(df['Arrival Delay in Minutes'].median(), inplace=True)
```

```
df = df.dropna()
```

Encoding categorical predictors

```
df = pd.get_dummies(df, columns=['Customer Type', 'Type of Travel', 'Class'], drop_first = True)
```

```
X = df.drop('satisfaction', axis=1)
y = df['satisfaction']
```

```
df.head()
```

	satisfaction	Age	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	Inflight wifi service	Inflight entertainment	Online support	...	On-board service	...
0	satisfied	65	265	0	0	0	2	2	4	2.0	...	3.0	...
1	satisfied	47	2464	0	0	0	3	0	2	2.0	...	4.0	...
2	satisfied	15	2138	0	0	0	3	2	0	2.0	...	3.0	...
3	satisfied	60	623	0	0	0	3	3	4	3.0	...	1.0	...
4	satisfied	70	354	0	0	0	3	4	3	4.0	...	2.0	...

5 rows × 21 columns

Splitting data into training and testing sets for unbiased model evaluation

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
```

```
test_size=0.2,
random_state=42)
```

```
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(73584, 22)
(18396, 22)
(73584,)
(18396,)
```

```
df['satisfaction'] = df['satisfaction'].map({
    'satisfied': 1,
    'dissatisfied': 0})
```

```
print(df['satisfaction'])
```

```
0      1
1      1
2      1
3      1
4      1
..
23610   0
23611   0
23612   0
23613   1
23614   1
Name: satisfaction, Length: 23615, dtype: int64
```

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (sta
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```
    LogisticRegression
```

```
LogisticRegression(max_iter=1000)
```

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
from sklearn.metrics import precision_score, recall_score, classification_report
import matplotlib.pyplot as plt
```

```
y_pred = model.predict(X_test)
```

```
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[1961  378]
 [ 382 2002]]
```

Interpretation of the matrix

True Negative(TN) = 1961, meaning that 1961 were correctly predicted as 'not satisfied' by the model.

True Positive(TP) = 2002, meaning that 2002 were correctly predicted as 'satisfied' by the model.

False Negative(FN) = 382, meaning that 382 were wrongly predicted to be 'not satisfied' when they are 'satisfied'.

False Positive(FP) = 378, meaning that 378 were wrongly predicted to be 'satisfied' when they are 'not satisfied'.

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Checking factors that most strongly influence passengers satisfaction.

```
coefficients = pd.DataFrame({
    'Feature': X_train.columns,
    'Coefficient': model.coef_[0]
})
coefficients = coefficients.sort_values(by='Coefficient', ascending=False)
print(coefficients)
```

	Feature	Coefficient
7	Inflight entertainment	0.870142
18	Class_Eco	0.599477
2	Seat comfort	0.506584
8	Online support	0.285996
19	Class_Eco Plus	0.265048
9	Ease of Online booking	0.176974
10	On-board service	0.175974
11	Leg room service	0.151101
13	Checkin service	0.055661
16	Departure Delay in Minutes	0.000728
1	Flight Distance	-0.000861
5	Gate location	-0.012306
17	Arrival Delay in Minutes	-0.012446
0	Age	-0.018409
6	Inflight wifi service	-0.053183
12	Baggage handling	-0.054368
14	Cleanliness	-0.058062
15	Online boarding	-0.124729
3	Departure/Arrival time convenient	-0.151697
4	Food and drink	-0.363828

From the above coefficient analysis, 'Inflight entertainment', 'Class', 'Seat comfort' and 'Food and drink' have the stongest influence on passengers' satisfaction.

```
odds_ratios = pd.DataFrame({
    'Feature': X_train.columns,
    'Coefficient': model.coef_[0],
    'Odds Ratio': np.exp(model.coef_[0])})
print(odds_ratios.sort_values(by='Odds Ratio', ascending=False))
```

	Feature	Coefficient	Odds Ratio
7	Inflight entertainment	0.870142	2.387251
18	Class_Eco	0.599477	1.821166
2	Seat comfort	0.506584	1.659613
8	Online support	0.285996	1.331088
19	Class_Eco Plus	0.265048	1.303494
9	Ease of Online booking	0.176974	1.193601
10	On-board service	0.175974	1.192408
11	Leg room service	0.151101	1.163115
13	Checkin service	0.055661	1.057239
16	Departure Delay in Minutes	0.000728	1.000728
1	Flight Distance	-0.000861	0.999140
5	Gate location	-0.012306	0.987770
17	Arrival Delay in Minutes	-0.012446	0.987631
0	Age	-0.018409	0.981759
6	Inflight wifi service	-0.053183	0.948207
12	Baggage handling	-0.054368	0.947083
14	Cleanliness	-0.058062	0.943592
15	Online boarding	-0.124729	0.882736
3	Departure/Arrival time convenient	-0.151697	0.859248
4	Food and drink	-0.363828	0.695011

A one-unit increase in the inflight entertainment rating multiplies the odds of satisfaction by approximately 2.39, holding other variables constant.

Inflight entertainment is the highest driver of satisfaction, therefore, investing on high quality entertainment systems and contents are necessary. There should be improvement in the seat cushioning and leg support.

Food and drink has the strongest negative influence on satisfaction, the meal quality, varieties and beverages options should be reviewed.

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Model limitations

There are a lot of predictors which affected the model, from the predicts, the model made more than 300 wrong predictions.

I will reduce the predictors in order to reduce noises and build a better model